Category Theory for Mathematical Physics — Exercise Sheet 14

Topological quantum field theory: a synthesis

Variant: Hints and answers

Conventions as in Lecture 14: time runs upward; a Frobenius algebra is $(A, \mu, \iota, \Delta, \varepsilon_A)$, drawn with the pieces of § 14.2 (the pants μ and Δ, the discs ι and ε_A); P is the flip of Vect, $w = \mu \circ \Delta \circ \iota$ the handle element, Σ_g the closed surface of genus $g \geq 0$, and $Z(\Sigma_g) = \varepsilon_A(w^g)$. Groups are multiplicative, with identity 1, and $[a, b] = aba^{-1}b^{-1}$. In this sheet $\beta = \varepsilon_A \circ \mu$ and $\gamma = \Delta \circ \iota$. In Exercise 4, n_0, n_1, n_2 count vertices, edges and faces, and Z_1, B_1 are spaces of chains (not partition functions or centres); in the starred exercise $N(x)$ is a count.

Exercise 1 (The Frobenius axioms as surfaces). (i) Draw as surfaces, glued from the pieces of Lecture 14, the associativity of μ, the unit law $\mu \circ (\iota \otimes \mathrm{id}) = \mathrm{id}$ and the Frobenius relation Equation (14.3). Which pictures give coassociativity and the counit law? Which diffeomorphism of the pair of pants gives commutativity?

(ii) Show that $(\mathrm{id} \otimes \beta) \circ (\gamma \otimes \mathrm{id}) = \mathrm{id}_A = (\beta \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \gamma)$, draw it as in Equation (14.1), and deduce that β is a nondegenerate pairing with $\beta(xy, z) = \beta(x, yz)$.

Hint. (i) Count the genus and the incoming and outgoing circles: two connected surfaces with the same counts are diffeomorphic relative to the boundary. (ii) $\gamma \otimes \mathrm{id} = (\Delta \otimes \mathrm{id}) \circ (\iota \otimes \mathrm{id})$; use one form of the Frobenius relation.

Answer. (i) Each side of associativity and of the Frobenius relation is a sphere with four holes, the unit law a cylinder; coassociativity and the counit law are the same pictures upside down; commutativity is the half-twist exchanging the legs. (ii) Frobenius, then the unit and counit laws; a v with $\beta(v, -) = 0$ is 0 by the snake.

Exercise 2 (The handle element). (i) Let $A = \mathbb{C}[x]/(x^2)$ with $\varepsilon_A(1) = 0$, $\varepsilon_A(x) = 1$. Show that β is nondegenerate, find $\Delta(1)$ and $\Delta(x)$ from $\Delta(a) = \sum_i a e_i \otimes e^i$, where $\{e_i\}$ and $\{e^i\}$ are dual bases, $\beta(e_i, e^j) = \delta_{ij}$, and compute w and $Z(\Sigma_g)$ for every g.

(ii) For any commutative Frobenius algebra show that $Z(T^2) = \varepsilon_A(w) = \dim A$: the case $\Sigma = S^1$ of Theorem 14.2(ii).

(iii) Let $A = \mathbb{C}[\mathbb{Z}_2]$ with $\varepsilon_A(g) = \delta_{g,1}$. Compute $Z(\Sigma_g)$. What does the counit $\varepsilon_A(f) = f(1)/|G|$ of Theorem 14.6 give instead?

Hint. (i) The dual basis of $\{1, x\}$ is $\{x, 1\}$. (ii) By Exercise 1(ii), γ is the copairing of β. (iii) The idempotents $(1 \pm s)/2$, and Example 14.5.

Answer. (i) $\Delta(1) = 1 \otimes x + x \otimes 1$, $\Delta(x) = x \otimes x$, $w = 2x$; $Z(\Sigma_1) = 2$ and $Z(\Sigma_g) = 0$ for $g \neq 1$. (ii) $\varepsilon_A(w) = \sum_i \beta(e_i, e^i) = \dim A$. (iii) 2^g; with $f(1)/2$, 2^{2g-1}, which is Theorem 14.6 for $\mathbb{Z}_2$.

Exercise 3 (Dijkgraaf–Witten for $\mathbb{Z}_2$ and S_3). Use

$$\pi_1(\Sigma_g) = \langle a_1, b_1, \ldots, a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] \rangle.$$

(i) For $G = \mathbb{Z}_2$, count $\#\,\mathrm{Hom}(\pi_1\Sigma_g, G)$, and check both expressions of Theorem 14.6.

(ii) For $G = S_3$, compute $Z(\Sigma_g)$ for $g = 0, 1, 2$ from the character sum, and $\#\,\mathrm{Hom}(\pi_1\Sigma_2, S_3)$. Check $g = 0$ and $g = 1$ against the count of homomorphisms directly.

Hint. (i) In an abelian group every commutator is 1. (ii) A homomorphism from $\mathbb{Z}^2$ is a pair of commuting elements; count them as $\sum_a |C(a)|$.

Answer. (i) 2^{2g} homomorphisms, $Z = 2^{2g-1} = 2 \cdot 2^{2g-2}$. (ii) $\frac{1}{6}$, 3, 81; 486; $1/|G|$ at $g = 0$ and $18/6 = 3$ commuting pairs per $|G|$ at $g = 1$.

Exercise 4 (The Levin–Wen count for $\mathrm{Vec}_{\mathbb{Z}_2}$). Let Γ be a connected trivalent graph in Σ_g whose faces are discs, with n_0 vertices, n_1 edges and n_2 faces, so that $n_0 - n_1 + n_2 = 2 - 2g$. The string net of $\mathrm{Vec}_{\mathbb{Z}_2}$ puts a qubit, a label 0 or 1, on each edge; $Q_v = 1$ when the labels at v add up to 0 mod 2, and $B_p = \frac{1}{2}(1 + X_p)$, where X_p adds ∂p, the boundary of the face p mod 2, to the labels (an edge along which ∂p runs twice is not flipped) (§ 14.4, with every $d = 1$ and every F-symbol 1).

(i) Let Z_1 be the closed $\mathbb{Z}_2$-chains (the label sets with $Q_v = 1$ at every v) and B_1 the sums of face boundaries. Show that the ground states are the functions on Z_1 that are constant on the cosets of B_1.

(ii) Show that $\dim Z_1 = n_1 - n_0 + 1$ and $\dim B_1 = n_2 - 1$ over $\mathbb{Z}_2$, and deduce that there are 2^{2g} ground states.

(iii) Compare the torus with Theorem 14.8(ii)–(iii) and with *Quantum Computing*. How many torus states does Theorem 14.8 give for Vec_{S_3}, and why is that not the $Z(T^2) = 3$ of Exercise 3?

Hint. (i) Q_v and B_p are commuting projectors, and the ground states have all of them equal to 1. (ii) Use connectedness to find the image of the boundary map from edges to vertices, and the kernel of the map from faces to edges.

Answer. (i) $Q_v = 1$ restricts to Z_1; $X_p\psi = \psi$ makes ψ invariant under adding ∂p. (ii) $\dim Z_1/B_1 = n_1 - n_0 - n_2 + 2 = 2g$. (iii) $4 = \operatorname{rank} \mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$, the toric code; 8 for Vec_{S_3}, a 3D state space, against the 2D partition function 3.

Exercise 5 (Turaev–Viro, Reshetikhin–Turaev and the anomaly). (i) Give $\dim Z_{\mathrm{TV},C}(T^2)$ for $C = \mathrm{Vec}_{\mathbb{Z}_2}$, Vec_{S_3}, Fibonacci and Ising.

(ii) For Fibonacci, Ising, $\mathrm{SU}(2)_2$, the toric code and $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$, find c mod 8 (Lecture 10) and say whether the gluing phase $e^{2\pi i c/8}$ of RT is 1.

(iii) Why is no Turaev–Viro theory equal to $\mathrm{RT}_{\mathrm{Fib}}$?

Hint. (i) Theorem 14.8(ii)–(iii), and Theorem 12.10 for the modular ones. (iii) The central charge of a centre.

Answer. (i) 4, 8, 4, 9. (ii) $\frac{14}{5}$, $\frac{1}{2}$, $\frac{3}{2}$, 0, 0: only the toric code and $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$. (iii) $c(\mathcal{Z}(C)) = 0$, while $c(\mathrm{Fib}) = \frac{14}{5}$.

Exercise 6 ⋆. *Mednykh's formula.* Let G be a finite group, K its conjugacy classes, and for $x \in G$ let

$$\mathcal{N}(x) = \#\{(a, b) \in G^2 : [a, b] = x\}.$$

You may use, for the irreducible characters, $1_K = \frac{|K|}{|G|} \sum_\chi \overline{\chi(K)}\, \chi$ (the indicator function of the class), the relation $\sum_K |K|\ |\chi(K)|^2 = |G|$, and $\chi * \psi = \delta_{\chi\psi}\, (|G|/\chi(1))\, \chi$ for the convolution of Theorem 14.6.

(i) Show that $\mathcal{N} = \sum_K \frac{|G|}{|K|}\, 1_{K^{-1}} * 1_K$.

(ii) Deduce that $\mathcal{N}(x) = |G| \sum_\chi \chi(x)/\chi(1)$.

(iii) Show that $\#\operatorname{Hom}(\pi_1 \Sigma_g, G) = \mathcal{N}^{*g}(1)$, and conclude that $\#\operatorname{Hom}(\pi_1 \Sigma_g, G)/|G| = \sum_\chi (|G|/\chi(1))^{2g-2}$.

Hint. (i) For fixed a, write $[a,b]=x$ as $ba^{-1}b^{-1}=a^{-1}x$; how many b conjugate a^{-1} to a given element of its class? (ii) $\chi(K^{-1})=\chi(K)$. (iii) Split $x_1\cdots x_g=1$ with $x_i=[a_i,b_i]$.

Answer. (i) The identity holds: for each a, count the b. (ii) $\mathcal{N}=|G|\sum_\chi \chi/\chi(1)$. (iii) $\mathcal{N}^{*g}(1)=|G|^{2g-1}\sum_\chi \chi(1)^{2-2g}$; dividing by $|G|$ gives the formula.